Maths Behind Transformers

ML Men Group
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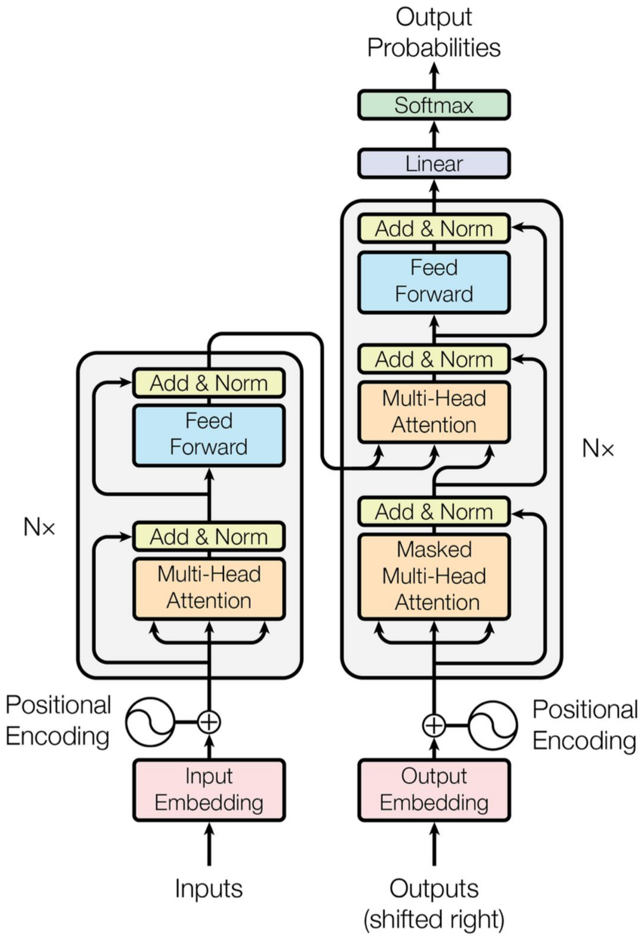
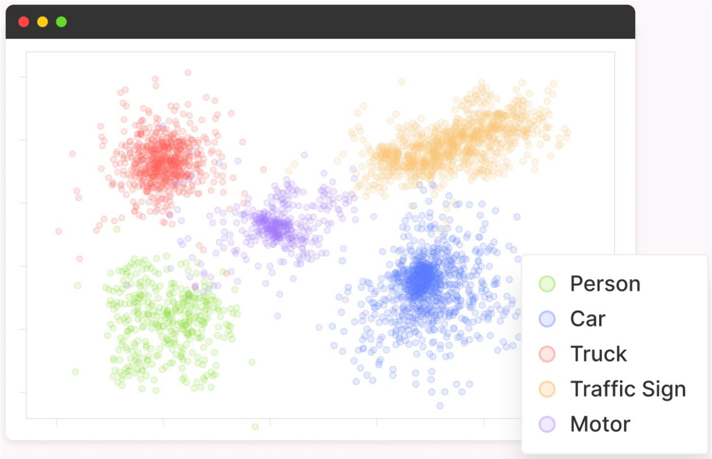
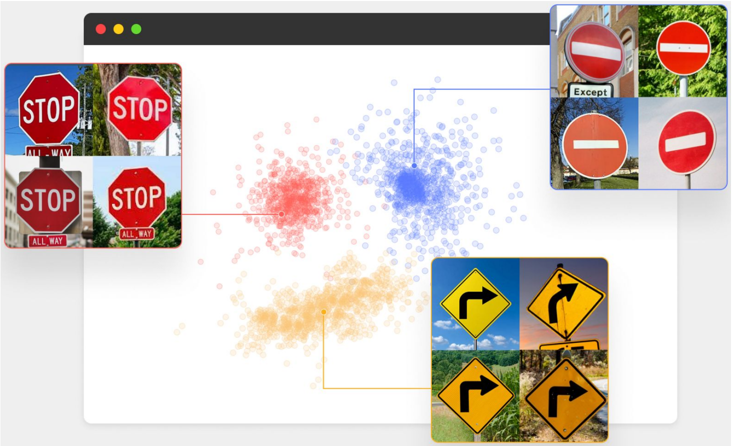


Figure 1: The Transformer - model architecture.

Embeddings

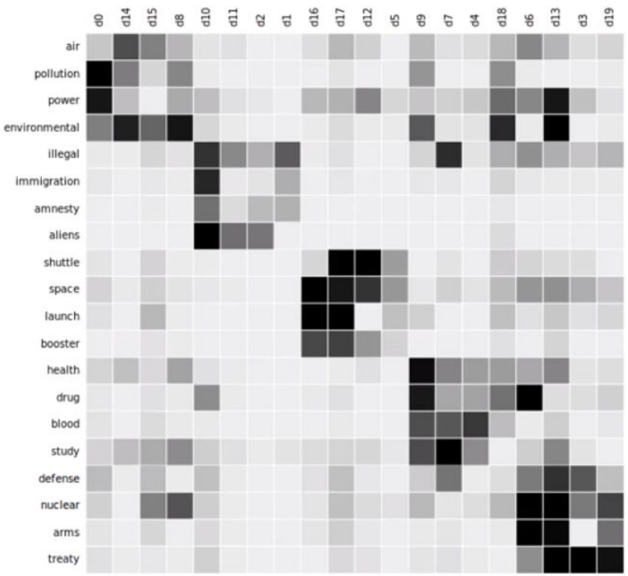
- To mathematically represent abstract concepts (words or images) so that computers can understand it.



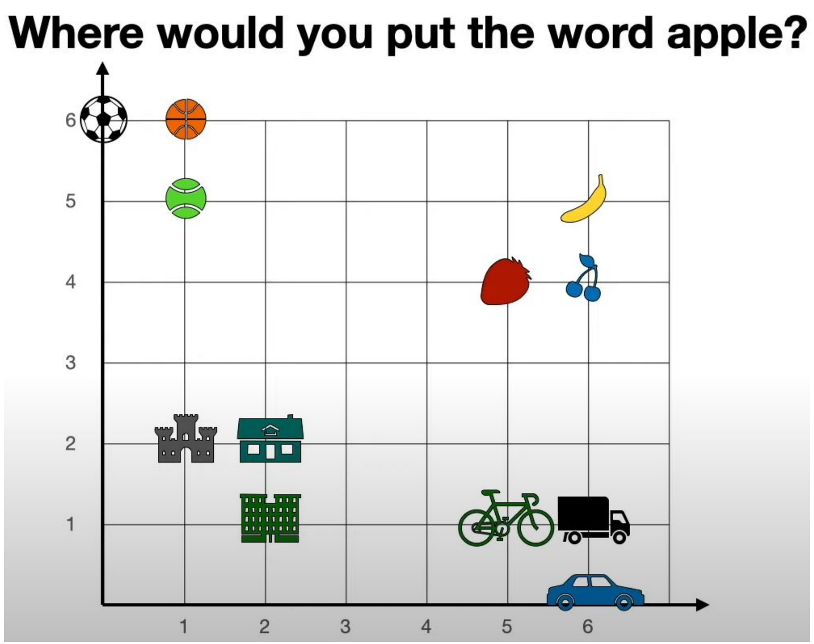


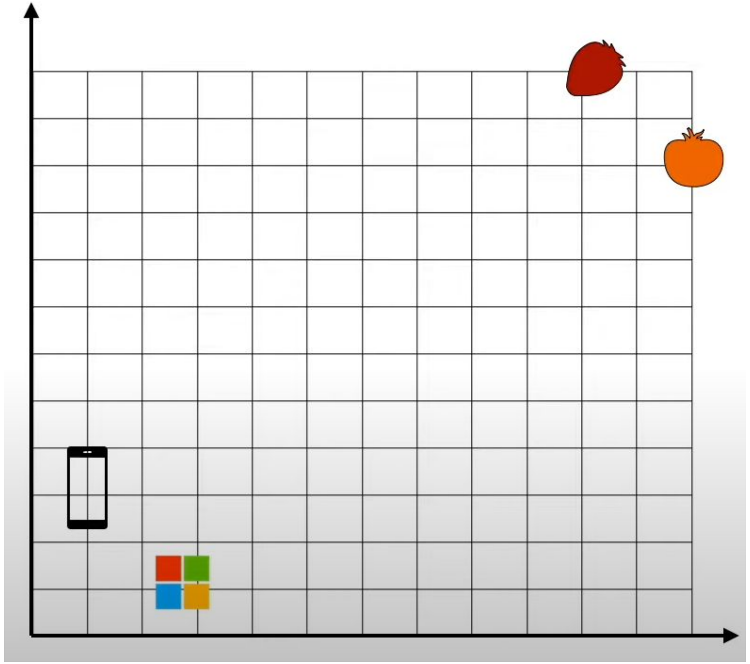
Embeddings

- Earlier methods to embed data was non-contextualized and fairly primitive. [1, 2]

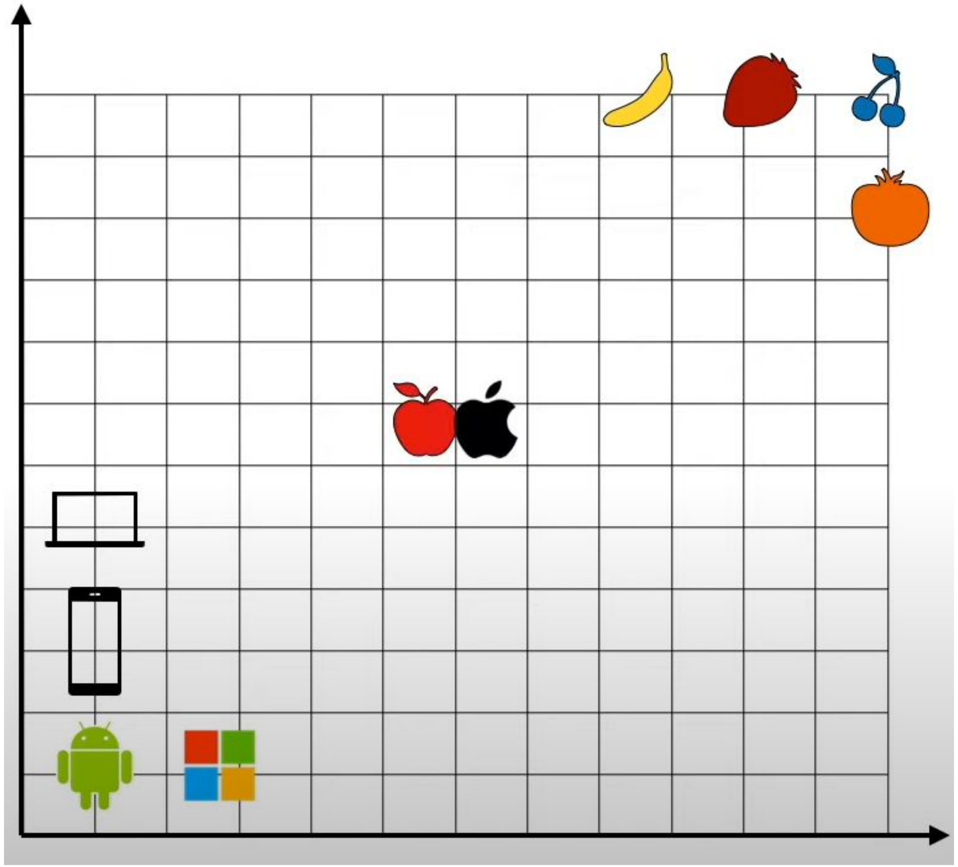


Embeddings

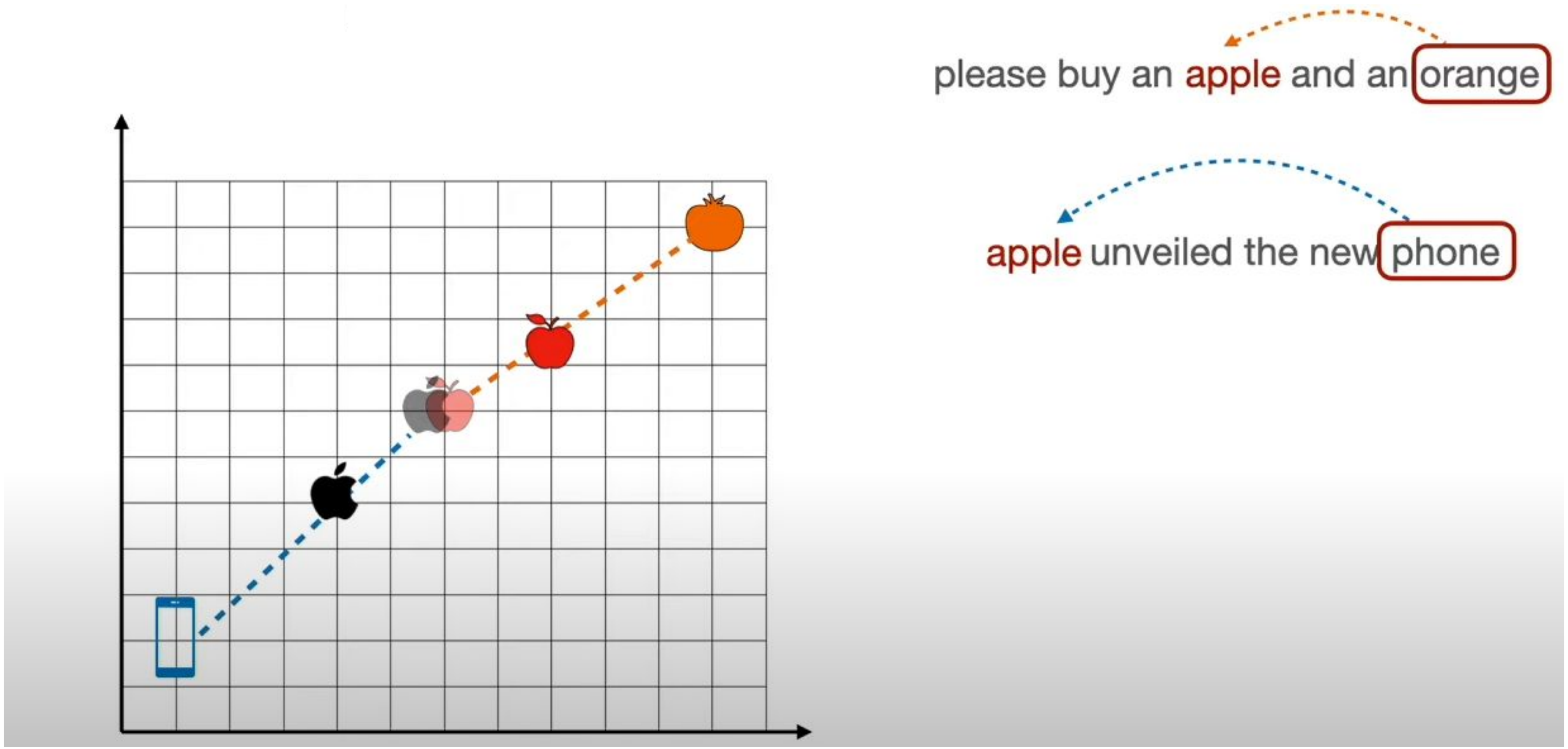




Embeddings

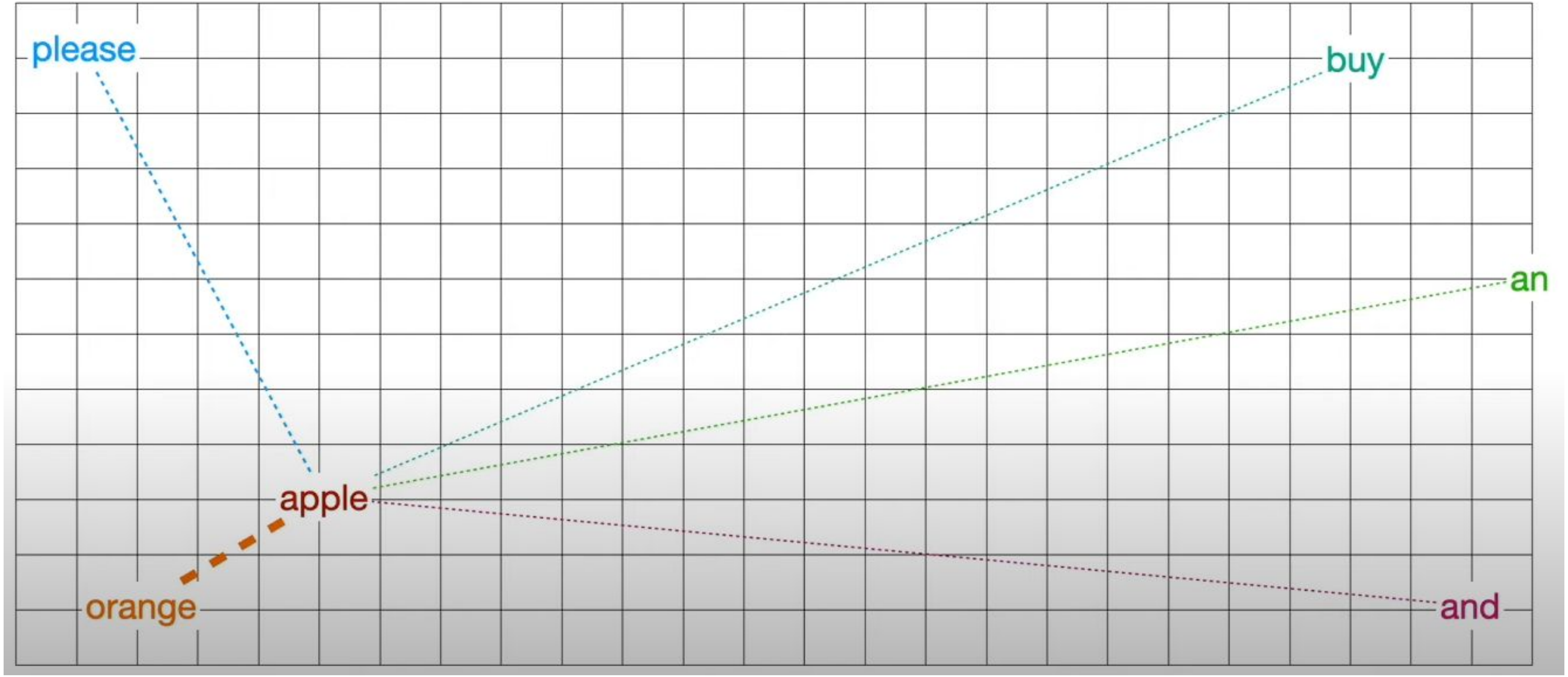


Attention


please buy an apple and an orange
apple unveiled the new phone

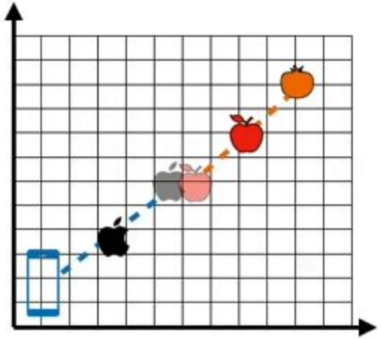
Attention

please buy an apple and an orange

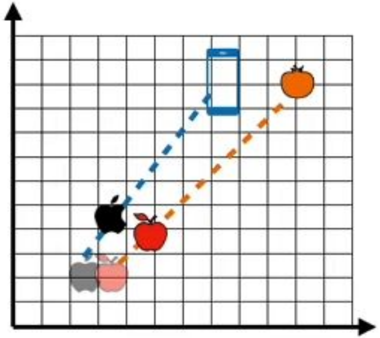


please
buy
an
apple
orange
and

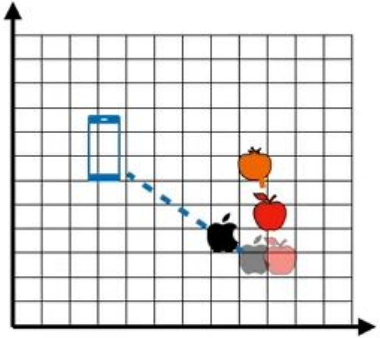
Multi-head Attention



Good



Bad

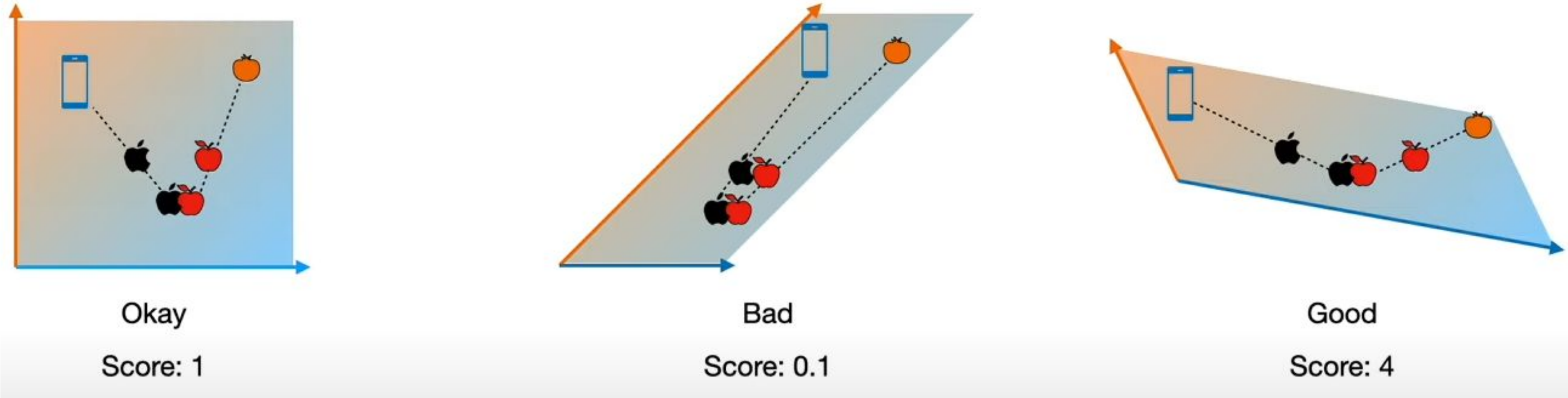


So-so

Problem: Building many embeddings is a lot of work!

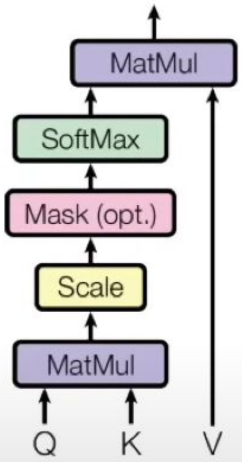
Solution: We'll build embeddings by modifying existing embeddings

Multi-head Attention



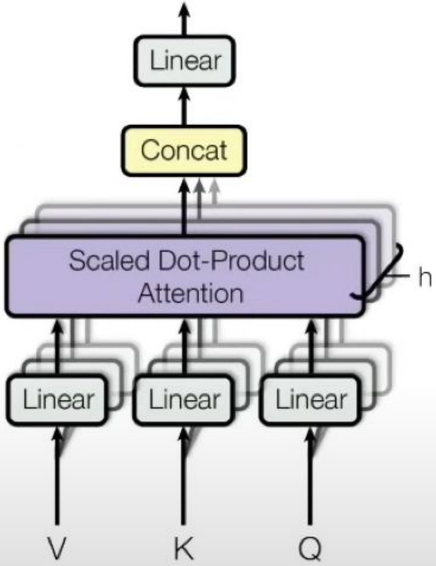
Keys, Queries & Values

Scaled Dot-Product Attention



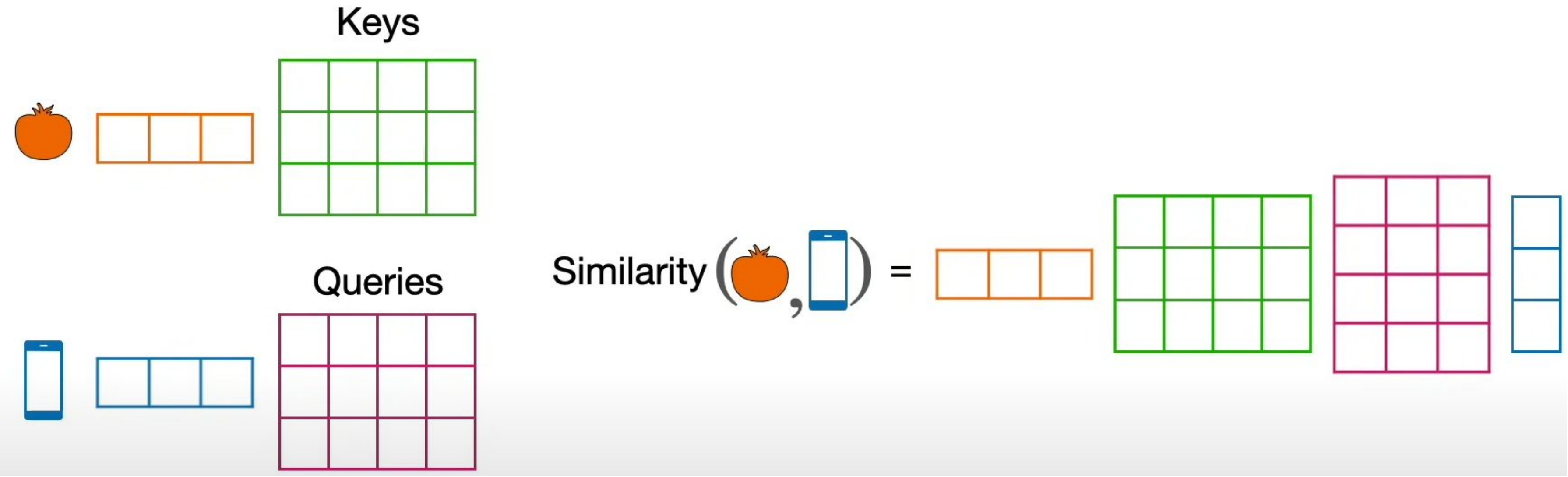
$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Multi-Head Attention

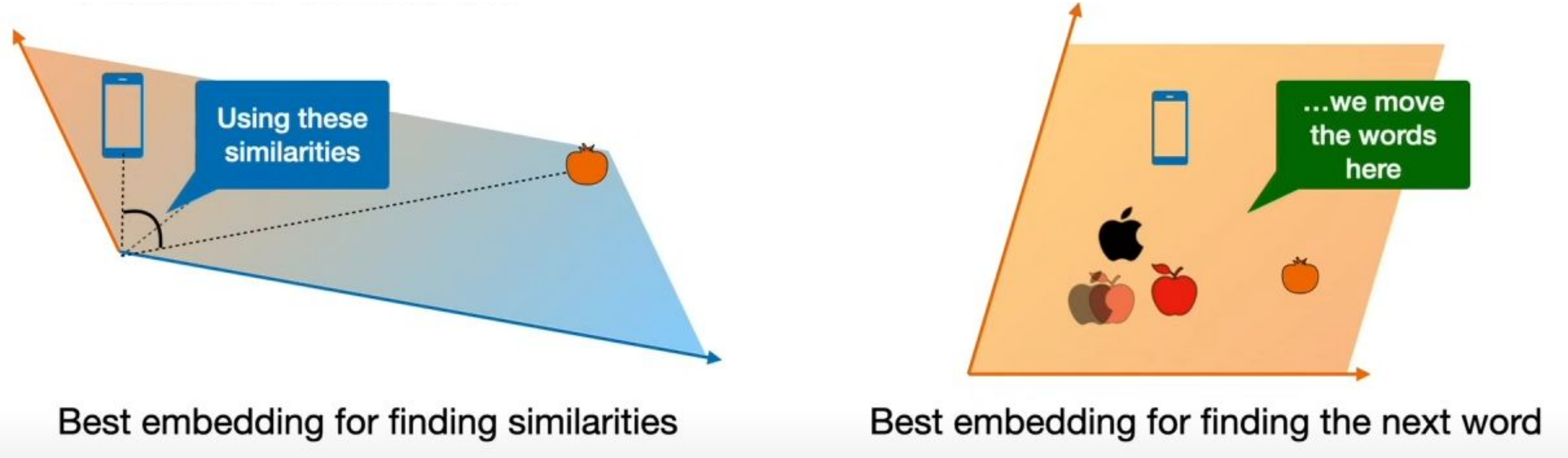


$$\mathrm{MultiHead}(Q,K,V) = \mathrm{Concat}(\mathrm{head_1}, ..., \mathrm{head_h})W^O$$
$$\text{where } \mathrm{head_i} = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Keys, Queries & Values



Keys, Queries & Values



References

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2. The Illustrated Word2vec
3. Large Language Models [Playlist]
4. Illustrated Guide to Transformers Neural Network: A step by step explanation
5. The Neuroscience of "Attention"
6. CS480/680 Lecture 19: Attention and Transformer Networks